

Practice Exam 2 (Challenge) · Answer Key

Chemistry Interim · 80 points · Unit 1: 48 points, Unit 2: 16 points, Unit 3: 16 points

How to use this key. This exam is marked strictly on detail. A symbol with the wrong charge, a number without its unit, or an explanation without the underlined key terms loses that point.

Part A · Multiple Choice (2 points each, 36 total: Unit 1 is questions 1-4, 7-10 and 13-15; Unit 2 is 5, 11, 16, 17; Unit 3 is 6, 12, 18)

Quick key: 1 D 2 C 3 A 4 B 5 B 6 C 7 A 8 D 9 B 10 C 11 D 12 A 13 C 14 D 15 B
16 C 17 A 18 D

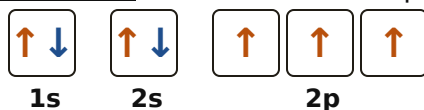
#	Ans	Why
1	D	Bohr's model. Bohr placed electrons in fixed energy levels to explain hydrogen's line spectrum.
2	C	Electrons move in fixed energy levels. Energy levels came later, from Bohr and the hydrogen spectrum. Watch the word NOT.
3	A	protons. Protons identify the element. Protons plus neutrons is the mass number.
4	B	the neutron. Electron 1897, nucleus 1911, proton about 1919, neutron 1932. A neutral particle is the hardest to detect.
5	B	F, Cl, Br. Down a group the radius increases: $F < Cl < Br$. The other three lists all decrease.
6	C	copper. Metallic bonding: a sea of delocalized electrons. In solid NaCl the ions are fixed and there are no free electrons.
7	A	23. Electrons = protons – charge = $26 - 3 = 23$. A positive charge means electrons were lost. 30 is the number of neutrons.
8	D	$^{16}_8\text{O}$ and $^{18}_8\text{O}$. Same protons (8), different neutrons (8 and 10). A is two different elements; B and C are an atom and its ion.
9	B	The two isotopes are present in nearly equal amounts. 79.90 is almost exactly halfway between 79 and 81, so the two are close to 50% each (actually 50.7% and 49.3%).
10	C	$1s^2 2s^2 2p^5 3s^1$. 10 electrons (neon), but one has moved to 3s while 2p is not full. The other three follow the filling order.
11	D	Mg^{2+}. All four have 10 electrons. Mg^{2+} has the most protons (12), so it pulls them in the most.
12	A	N_2O_3, dinitrogen trioxide. B needs a Roman numeral: copper(II) chloride. C is ionic, so no prefix: calcium chloride. D is sodium sulfate; sulfide is S^{2-} .
13	C	$[\text{Ar}] 4s^2 3d^{10} 4p^4$. [Ar] is 18 electrons; $18 + 2 + 10 + 4 = 34$. A forgets the 3d electrons; B uses the noble gas after selenium, not before it.
14	D	$n = 2$ to $n = 1$. The levels are not evenly spaced: the gap between $n = 1$ and $n = 2$ is by far the largest ($1.63 \times 10^{-18} \text{ J}$, against $0.41 \times 10^{-18} \text{ J}$ for 4 to 2). Counting the number of levels crossed gives the wrong answer.
15	B	^1_1H. Mass numbers: $2 + 2 = 3 + 1$. Atomic numbers: $1 + 1 = 1 + 1$. Mass 1 and charge 1 is a proton. A neutron would leave the atomic numbers unbalanced.

16	C	13. The huge jump comes after the third electron, so there are 3 valence electrons: group 13 (this is aluminum).
17	A	Chlorine has a greater nuclear charge with about the same shielding, so its outer electrons are held more tightly. Both are in period 3: same number of levels, same shielding, but 17 protons against 11. B, C and D are all false for these two atoms.
18	D	CF₄. C-F bonds are polar (ΔEN 1.5), but the tetrahedral shape with four identical atoms cancels the dipoles. Cl ₂ has a nonpolar bond.

Part B - Free Response (44 points)

19. Find the mistakes (8 points)

A. It breaks **Hund's rule**: two electrons are paired in one 2p orbital while another 2p orbital is empty. Correct:



B. The **4s sublevel fills before 3d** (Aufbau principle), and it was left out. Correct: **1s² 2s² 2p⁶ 3s² 3p⁶ 4s² 3d⁶** (26 electrons).

C. Two mistakes. (1) The mass number is **protons + neutrons** = 12 + 13 = 25, not 13. (2) The charge is protons – electrons = 12 – 10 = **2+**, not 2–: losing electrons makes a positive ion.

Correct: **²⁵₁₂Mg²⁺**

D. The core must be the **previous noble gas**. Argon (18 electrons) comes after chlorine (17). Correct: **[Ne] 3s² 3p⁵**.

2 points each: 1 for naming the mistake(s), 1 for the correct answer. In C both mistakes are needed for the first point.

20. Careful calculations (10 points)

(a) 1 point per fully correct row.

Species	Protons	Neutrons	Electrons	How
³¹₁₅P^{3–}	15	16	18	31 – 15; 3– means 3 electrons gained
⁶⁵₂₉Cu²⁺	29	36	27	65 – 29; 2+ means 2 electrons lost
¹³⁷₅₆Ba²⁺	56	81	54	137 – 56; 2+ means 2 electrons lost

(b) Percent of Cu-65 = 100 – 69.15 = 30.85%.

$$(62.930 \times 0.6915) + (m \times 0.3085) = 63.546$$

$$43.516 + 0.3085m = 63.546, \text{ so } m = 20.030 \div 0.3085 = \mathbf{64.93 \text{ amu}}$$

1 point: 30.85% found by subtraction. 1 point: equation with fractions, not percents. 1 point: answer with unit (accept 64.92 to 64.93).

(c) Let x = fraction of Li-6, so $(1 - x)$ = fraction of Li-7.

$$6.015x + 7.016(1 - x) = 6.940$$

$$7.016 - 1.001x = 6.940, \text{ so } x = 0.076 \div 1.001 = 0.0759$$

Li-6: 7.6% · Li-7: 92.4%

1 point: set-up with x and $(1 - x)$. 1 point: algebra. 1 point: both percents. Check: 6.940 is very close to 7, so Li-7 must be the large share. Swapping the two percents loses the last point.

(d) $2\text{}^1_1\text{H}$ (two protons). Mass numbers: $3 + 3 = 4 + 2(1)$. Atomic numbers: $2 + 2 = 2 + 2(1)$.

21. Light and spectra (8 points)

(a) The electron starts in the **ground state** (its lowest energy level). It **absorbs** energy from the electricity and jumps to a higher level; the atom is now in an **excited state**. The electron then falls back to a lower level (from $n = 3$ to $n = 2$ for the red line) and **emits** the energy difference as one **photon** of red light.

1 point: absorbs energy and moves up (ground to excited). 1 point: falls back to a lower level. 1 point: emits a photon whose energy equals the difference between the levels. No credit for saying light is emitted as the electron moves up.

(b) The energy levels are **quantized**: the electron can only be in certain levels, so only **certain energy drops** are possible. Each drop gives one photon energy, so only certain wavelengths (colors) appear.

1 point: fixed (quantized) energy levels. 1 point: only certain energy differences, so only certain colors.

(c) **Mercury and sodium are present.** All four mercury lines (405, 436, 546, 578 nm) and all four sodium lines (498, 569, 589, 616 nm) are in the unknown. **Hydrogen is absent:** its lines at 486 and 656 nm are missing (410 and 434 are only close to mercury's 405 and 436). **Neon is absent:** only its 616 nm line matches (that line is sodium's); 540, 603, 627, 640, 660 and 693 nm are missing.

1 point: mercury and sodium, with all lines matching. 1 point: hydrogen rejected with a named missing line. 1 point: neon rejected with a named missing line (one shared line is not enough).

22. Periodic trends (8 points)

(a) **K < Na < S < Cl** (419, 496, 1000 and 1256 kJ/mol)

2 points for the full order; 1 point if exactly one pair is swapped.

(b) Both atoms have their outer electrons in the **same energy level** ($n = 2$) with the **same shielding** (the $1s^2$ core). Fluorine has **9 protons** against lithium's 3, so its greater nuclear charge pulls the outer electrons much closer.

1 point: same level / same shielding. 1 point: greater nuclear charge pulls the electrons in.

(c) **S²⁻ > Cl⁻ > K⁺ > Ca²⁺.** They are **isoelectronic**: each has 18 electrons. The number of **protons increases** (16, 17, 19, 20), so the same electrons are pulled in more strongly and the ion gets smaller.

1 point: order. 1 point: same electrons, more protons.

(d) **2 valence electrons; ion X²⁺.** The ionization energy rises steadily for the first two electrons, then makes a **very large jump** (1451 to 7733 kJ/mol) for the third, which must come from an **inner (core) level**. So X loses two electrons easily (it is magnesium, group 2).

1 point: 2 valence electrons and 2+. 1 point: the large jump after the second electron.

23. Chemical bonding (10 points)

(a) 1 point per row.

Atoms	ΔEN	Type of bond
Mg and O	$3.5 - 1.2 = 2.3$	ionic (metal + non-metal)
C and Cl	$3.0 - 2.5 = 0.5$	polar covalent (0.5 is the start of the polar range)

(b) 1 point per row.

Name	Formula	Check
aluminum oxide	Al₂O₃	Al ³⁺ and O ²⁻ : $2(3+) + 3(2-) = 0$
iron(II) nitrate	Fe(NO₃)₂	Fe ²⁺ and NO ₃ ⁻ ; parentheses around the polyatomic ion
diphosphorus pentoxide	P ₂ O ₅	two non-metals: prefixes
ammonium sulfate	(NH ₄) ₂ SO ₄	two polyatomic ions; no prefixes



CO₂ is **linear** (2 electron domains, no lone pairs on C), so its two equal C=O dipoles point in opposite directions and **cancel**: nonpolar. H₂O is **bent** (2 lone pairs on O), so its two O-H dipoles do not cancel: **water is the polar one**.

1 point: CO₂ with two double bonds and 2 lone pairs on each O. 1 point: H₂O with 2 lone pairs on O. 1 point: linear, dipoles cancel. 1 point: bent, dipoles do not cancel. A missing lone pair loses that structure's point.